

Phronesis

An AI vehicle diagnostic and advisory assistant for Uganda. Status as of 12 September 2026 — 50 commits, branch frontend-v2-migration.

What it does today

A driver describes a problem in plain language, or plugs in an OBD reader. Phronesis answers in her own voice, works out what is likely wrong, prices the repair split into parts and labour, and shows real garages nearby. Everything on every screen now comes from a real source — no screen runs on mock data.

The eleven screens

Screen	What it does	Data source
Loading	Wordmark with the avatar as the letter O	—
Welcome	She introduces herself, aloud	TTS
Onboarding 1	Owner or buyer — the branch the whole app turns on	local
Onboarding 2	Name, email, password in one form	Firebase Auth
Onboarding 3	Pair an OBD reader (optional, skippable)	Bluetooth LE
Home	Chat. Avatar centred, then docked as the microphone	POST /api/chat
Diagnosis	A real report, fault marked on the car	POST /api/diagnosis
Solutions	Repair options priced parts vs labour	saved diagnosis
Maps	Real garages on dark tiles	OpenStreetMap
Account	You, your car, alerts, voice, your data	/api/car-profile etc.
Style guide	Every atom, plus Principles, which measures itself	live DOM

The stack

Layer	Choice	Why
Frontend	React 19 · TypeScript · Vite 8	13,094 lines across 66 files
Styling	Plain CSS on a token layer + Tailwind v4	components, not utility classes, so controls cannot drift
Backend	Express · TypeScript, port 3001	1,595 lines; an I/O-bound LLM proxy
AI	Gemini, with Claude as fallback	streamed over SSE
Voice	Microsoft neural via msedge-tts	four other providers were tried and each failed differently
Auth	Firebase, Bearer ID token to the server	—
Map	Leaflet on Esri dark tiles	40kB, no key, no vector-tile account
Native	Capacitor — Android shell builds	10.4MB debug APK, minSdk 24

One language end to end. C# and R were considered and rejected: the backend spends 95% of each request waiting on someone else's server, so raw speed buys nothing, and R is a statistics language with no place in a production app. Python earns a place later, as a separate service, when the custom model is trained.

The design system

- **Hierarchy is brightness, not colour.** Three ink levels, all measured against every surface they sit on.
- **Colour means one thing.** Red and amber appear only where safety is at stake. There is no green — "nothing wrong" is the app's default state and reads as neutral.

- **The accent is light, never a fill.** It marks focus, an active tab, a route on the map. It never sits behind a label.
- **Press is light.** Every clickable surface warms from the point of contact, driven by one delegated listener.
- **Scale is fluid.** The root font size runs 15px to 19px with the viewport, and everything is relative to it.

Accessibility, measured rather than claimed

The style guide opens with a Principles section that reads the live page and reports pass or fail. It scores **11/11** and has already caught bugs no one was looking for.

Check	Was	Now
Dimmest text contrast	2.07–3.44:1 — failed AA everywhere	5.5:1
Destructive text	4.37:1	5.15:1
Smallest text	8px	11px floor, enforced
Star rating targets	14 × 14px	36 × 44px
Route changes	silent — no title, no focus, no announcement	all three
Skip link	none	first tab stop on every page

Verified

tsc && vite build and oxlint clean at every commit, 62 files. All 11 routes render with no console errors at 390, 834, 1280 and 1600px — zero overlaps, zero overflow, no horizontal scroll. The OBD trouble-code decoder passes 14 spec-derived cases, including the multi-frame reply that naive parsers silently truncate.

Not verified — needs hardware

No physical ELM327 and no Android handset were available. Unproven: that a real dongle's GATT profile matches, that the ELM327 handshake completes against real firmware, and that a car answers the probe. Written against the specification and the common clone profiles — an educated implementation, not a tested one.

What is left

Open item	Kind	Blocks
Chat history is saved but never read back	gap	"your past conversations" does not exist
Mechanic prices and vehicle specs	data	Solutions is only as trustworthy as its numbers
Account deletion vs. reviews left for mechanics	policy	the confirm dialog admits it is unsettled
Currency and locale	decision	hard-coded to UGX in one file
XSPACE display licence	legal	Orbitron is standing in
Avatar video clip provenance	legal	marked placeholder, unknown source
Pre-car journey (Discover, Compare)	phase 3	needs a vehicle-price corpus first

The one hardware decision

Buy an ELM327 that says BLE or Bluetooth 4.0. Both available APIs are Bluetooth Low Energy only. Most cheap dongles are Bluetooth Classic and are invisible to a phone app entirely — on iOS, unreachable by anything without Apple MFi certification. This is a purchasing decision before it is a code one, and the app now says so on the pairing screen.